

PROTEIN INHIBITORS OF PROTEINASES

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CONTENTS

PERSPECTIVES AND SUMMARY	593
INTRODUCTION	595
<i>Limited Knowledge of Physiological Function</i>	595
<i>Nomenclature Problems</i>	596
α_2 MACROGLOBULINS	598
INHIBITORS WITH CLASS-SPECIFIC REACTIVE SITES	599
<i>Inhibitors of Carboxyl, Metallo and Sulfhydryl Proteinases</i>	599
<i>Inhibitors of Serine Proteinases</i>	601
<i>The Standard Mechanism</i>	601
<i>The nature of the reactive site</i>	604
<i>Multiple reactive sites on a single polypeptide chain</i>	608
<i>The Inhibitor Families</i>	610
<i>The pancreatic trypsin inhibitor (Kunitz) family</i>	610
<i>Pancreatic secretory trypsin inhibitor (Kazal) family</i>	613
<i>Streptomyces subtilisin inhibitor family</i>	616
<i>Bowman-Berk inhibitor family</i>	616
<i>Soybean trypsin inhibitor (Kunitz) family</i>	617
<i>Other inhibitor families</i>	618
<i>Inhibitors in mammalian blood plasma</i>	619

PERSPECTIVES AND SUMMARY

Protein inhibitors of proteinases are ubiquitous. They are present in multiple forms in numerous tissues of animals and plants as well as in microorganisms. Their gross physiological function is the prevention of unwanted proteolysis, but detailed physiological function has been only rarely elucidated. With the exception of macroglobulins, which "inhibit" proteinases of all classes, individual protein inhibitors inhibit only proteinases belonging to a single mechanistic class. Of these the most studied are the inhibitors of serine proteinases. Most of these interact with the proteinases according to a common mechanism. In each inhibitor molecule there exists on the surface one peptide bond, the reactive site, which combines with the enzyme

in a substrate-like manner and serves as a substrate for the enzyme. However, while k_{cat}/K_m for this interaction is large, both k_{cat} and K_m are, at neutral pH, very small.

In the stable enzyme-inhibitor complex the reactive site peptide bond of the inhibitor is still intact, but the carbonyl carbon is no longer trigonal. It is appreciably pyramidalized by the interaction of its oxygen with the "oxyanion hole" of the enzyme (not, as believed earlier, by the formation of a fourth bond with O γ oxygen of the catalytic seryl residue of the enzyme). The conformation of residues surrounding the reactive site peptide bond is that of an optimal substrate. Reactive sites are very rigid. Upon enzyme-inhibitor interaction conformational changes in either partner are minimal; it is almost a classical lock and key interaction.

The known inhibitors are not all homologous but consist of several families, which must have arisen by convergent rather than divergent evolution.

A striking characteristic of inhibitors is the presence of several homologous reactive sites on the same polypeptide chain, which arise from gene elongation by repeated duplication. In the case of the Bowman-Birk inhibitor family this gives rise to two homology regions, which are interconnected by interhomology region disulfide bridges. In many other cases (ovomucoids, ovinhibitors-proteinase inhibitors from avian serum, dog submandibular inhibitor, inter- α -trypsin inhibitors from mammalian serum) the gene doubling gives rise to several (two, three, or six) tandem domains with all disulfide bridges intradomain and only short connecting peptides connecting the domains. Chelonianin from turtle egg white consists of two tandem domains on the same polypeptide chain, but the two domains belong to two different inhibitor families.

The inhibitors are exceptional among proteins since they tend to retain their inhibitory activity upon replacement of their reactive site residue (P_1) by another residue. In some cases such a substitution leads to a predictable change in inhibitory activity, e.g. Arg⁶³ $\rightarrow$ Trp⁶³ in soybean trypsin inhibitor (Kunitz) leads to the conversion of a trypsin inhibitor to a chymotrypsin inhibitor. Such changes, which can be carried out in the laboratory, also occur during inhibitor evolution. Thus within each inhibitor family the P_1 residue is not conserved, but changes frequently, often resulting in changes in inhibitory specificity. In some cases (the third domains of avian ovomucoids) this residue is the most variable residue in the entire domain. This hypervariability of reactive sites during evolution stands in sharp contrast to the evolution of other proteins, where active sites are strongly conserved. However, in most other proteins, substitutions at active sites lead to the loss of activity.

INTRODUCTION

Protein inhibitors of proteinases have not been reviewed here before. However, a book (1) and numerous reviews on this topic (2–13) have appeared elsewhere. Two international conferences on inhibitors were held (14, 15), and inhibitors were the main topic of the 23rd meeting on Protides of Biological Fluids (16). Extensive discussions of inhibitors are included in many books and symposia on the biological role of proteinases (17–22). The literature of this field is huge, with about 500 titles appearing per year. Limitations of our competence and of space led us to deemphasize in this review descriptions of partially characterized inhibitors, and of extensive studies on the physiological role and possible medical applications of protein inhibitors. We are also largely omitting the extensive and highly controversial literature on the mechanism of action of large inhibitors in mammalian blood. Because the mechanism of action of small inhibitors of serine proteinases is by now rather well known, we focus instead on the newer, extensive information on amino acid sequences of a large number of inhibitors.

While protein inhibitors of enzymes other than proteinases are well known [e.g. inhibitors of α -amylases (23), deoxyribonuclease I (24), phospholipase A (25), and protein kinases (26)], such inhibitors seem relatively rare compared to protein proteinase inhibitors. The reason for this seems easy to guess at: proteins are substrates for proteinases. All of the presently accepted and proposed mechanisms of action of protein proteinase inhibitors involve either a requirement for proteolysis or for action as a substrate analogue.

Limited Knowledge of Physiological Function

The physiological function of inhibitors in the broad sense is clear—elimination of unwanted proteolysis. However, detailed functions are not clear. It is generally agreed that the presence of secretory trypsin inhibitors (Kazal) in the pancreas of vertebrates prevents premature activation of trypsinogen and, in turn, of other pancreatic zymogens. The predisposition to emphysema found in individuals with genetically determined low levels of α_1 -proteinase inhibitor strongly suggests protection against excessive proteolysis of lung tissues, but, while leukoeytic elastase is the frequently proposed target enzyme for α_1 -proteinase inhibitor, that assignment is already somewhat controversial (27). It is generally agreed that the presence of large numbers of proteinase inhibitors in massive amounts in mammalian plasma serves to delimit blood clotting (28) and other proteolytic cascade

processes, such as hormone production and elimination and complement fixation (29). However, the precise physiological involvement of inhibitors in these processes is not known.

A good case has been made that many proteinase inhibitors in plants serve a defensive function against insect infestation by inhibition of insect proteinases. Ryan demonstrated that wounding of potato leaves leads to great increases in the level of a polysaccharide hormone, proteinase inhibitor-inducing factor (PIIF), which in turn leads to huge increases in inhibitor levels (30, 31). Many other exciting studies on the role of inhibitors are reported but in most cases the results, while suggestive, are far from unequivocal. In some cases, added research complicates rather than supports the naive original assignments. Thus an important role was at first envisaged for acrosin inhibitors in the process of sperm capacitation, but more recent work casts doubt on this conclusion (32).

For other inhibitors the situation is far worse, mainly because proteinase inhibitors are most frequently discovered and isolated by noting their ability to inhibit some readily available proteinase, e.g. bovine trypsin, rather than physiological target enzyme. They thus become proteins in search of a function, rather than proteins isolated to account for a previously discovered biological function. Because most laboratories used only bovine trypsin as the initial test enzyme, an opinion formed that most proteinase inhibitors are inhibitors of trypsin. As chymotrypsin, elastase, and subtilisin were added as screening tools, many inhibitors of these enzymes were discovered. None of these are likely to be true target enzymes for most of the isolated inhibitors. The lack of knowledge of true target enzymes for the majority of inhibitors is now one of the major stumbling blocks in the understanding of inhibitor evolution and of inhibitory specificity. It appears to us quite likely that the target enzyme problem can be best solved for intracellular proteinases and for their cognate inhibitors (33, 34).

Nomenclature Problems

In a review published nine years ago we deplored the confusion in inhibitor nomenclature (2). Since then a great deal about inhibitors has been learned, a huge number of new inhibitors discovered, and the confusion grows worse. Newly discovered inhibitors are usually named according to the biological source and the inhibited enzyme [e.g. S-SI, *Streptomyces* subtilisin inhibitor (35); plasminostreptin, plasmin inhibitor from *Streptomyces* (36)]. Both procedures cause confusion: the biological source because inhibitors belonging to more than one family may be present in the same source. In such cases we have suggested the use of the last name of the initial

discoverer as a distinguishing method (e.g. soybean trypsin inhibitor (Kunitz) and soybean trypsin inhibitor (Bowman-Birk); pancreatic trypsin inhibitor (Kunitz) and pancreatic secretory inhibitor (Kazal)]. However, this is not sufficient. In soybeans there exist many inhibitors of the Bowman-Birk family and these are now simply designated by letters and numbers (37). The Kunitz inhibitor from pancreas is not organ-specific but occurs in many bovine organs and most recently has been assigned to mast cells (38). The recent finding that the inhibitor in chicken plasma and chicken ovomucoid probably have the same amino acid sequence poses a serious problem (39, 40). The two inhibitors should probably be assigned the same name, but the well-established name, ovomucoid (41), seems grossly inappropriate for a plasma inhibitor. Naming after biological source clearly produces confusion and occasionally leads to the wasted effort of characterizing the same inhibitor several times. Even greater problems stem from the use of the inhibited enzyme as a part of the name. First, the discoverer usually does not know the true target enzyme or the enzyme that is most strongly inhibited. Thus α_1 -proteinase inhibitor was first named α_1 -anti-trypsin, when in fact many workers suspect that trypsin is not an important target enzyme for this inhibitor nor, among the many enzymes it inhibits, is trypsin most rapidly or strongly inhibited (42). Yet, because of the name, most studies of enzyme interactions with this inhibitor focus on trypsin. The suggestion to change the name to α_1 -proteinase inhibitor seems to have only limited acceptance. Another objection to specifying the inhibited enzyme as part of the name of the inhibitor is the recently discovered phenomenon of reactive site variability (37, 43–45). Frequently, highly homologous inhibitors from closely related species inhibit different enzymes. When this is the case, a search for a homologue may well fail if one is looking, for example, for trypsin inhibition, and the closely related homologue is now an elastase inhibitor (e.g. chicken and bobwhite quail ovomucoids).

Another strategy, which seemed superior, was to assign to inhibitors totally trivial names such as Trasylol[®], Bayer's trademark for pancreatic trypsin inhibitor (Kunitz), or ones that indicate only the source and inhibition but do not specify the enzyme, such as ovomucoid and ovomucoid inhibitor.

We suggest that a rational system of nomenclature should involve first assigning the inhibitor to a family, then specifying the number of domains (if more than one) and the species from which it is isolated. The additional designators, such as its tissue source and the enzyme inhibited should be added last, since they are most likely to be confusing. However, as the list of families is still quite preliminary and since assignment to a family requires sequencing, introduction of the new nomenclature is still premature.

α_2 MACROGLOBULINS

The term protein inhibitor of proteinases could in principle denote any protein that diminishes the enzymatic activity of a proteinase. The most bizarre examples, such as proteinases that "inhibit" other proteinases by digesting them, fortunately are not called proteinase inhibitors. However, two entirely dissimilar groups of proteins are called proteinase inhibitors. One of them consists of the macroglobulins, high-molecular-weight proteins apparently present in the plasma of all mammals, often in multiple forms. They are exemplified by α_2 human macroglobulin, often abbreviated as α_2 M, which has a molecular weight of 720,000 and consists of four apparently identical polypeptide chains. It dissociates into pairs of chains upon denaturation and into single chains upon reduction of disulfide bridges (46, 47).

When α_2 M combines with proteinases, only the proteolytic activity toward large protein substrates is decreased or eliminated. The activity toward specific burst substrates and small synthetic substrates is unimpeded (48). Strikingly, the α_2 M-proteinase complexes can still be inhibited by small protein proteinase inhibitors, such as pancreatic trypsin inhibitor (Kunitz) or pancreatic secretory trypsin inhibitor (Kazal), but not by larger ones such as soybean trypsin inhibitor (Kunitz) (49). Clearly, in the α_2 M complexes, the active site of the proteinase is quite open; it is the access of substrates and inhibitors to this site that is sterically hindered. The other remarkable feature of macroglobulins is their ability to bind a wide variety of proteinases belonging to all four mechanistic classes (50). Indeed, it is relatively rare to find endoproteinases that do not form complexes with macroglobulins. The above facts gave rise to the "trap" mechanism of macroglobulin action (51). The proteinase hydrolyzes one or more particularly susceptible peptide bonds in α_2 M, and triggers a conformational change in α_2 M which traps the enzyme molecule. Only molecules of very high molecular weight can act by this mechanism. To account for the broad specificity the critical peptide segment must be rather long since it must contain peptide bonds matching the specificities of the various proteinases. Evidence has been provided for specific proteolytic cleavages upon complex formation.

Much research in vivo and in vitro compares the importance of α_2 M with that of other inhibitors present in blood. Upon addition of a small quantity of enzyme to a mixture of α_2 M and another inhibitor, complexes with both inhibitors are formed in a ratio that depends upon the relative values of the second order rate constants for association. However, the complex with a typical inhibitor dissociates, albeit slowly, while the α_2 M complex is "irre-

versible" and does not. The result in an in vitro experiment is that all of the proteinase is ultimately transferred to an $\alpha_2\text{M}$ complex, unless there is so much enzyme present that $\alpha_2\text{M}$ is saturated. In vivo this effect is greatly exaggerated, since $\alpha_2\text{M}$ proteinase complexes are cleared very rapidly; the half-life is about 10 min in man (52). Thus, essentially all injected proteinases are cleared by the $\alpha_2\text{M}$ pathway. Other serine proteinase inhibitors are only transiently involved in transferring the proteinases to be cleared. There is considerable interest in understanding the signals leading to the more rapid clearance of complexed as opposed to free $\alpha_2\text{M}$. The view presented above is an idealization and probably does not apply to all serine proteinases.

The research on macroglobulins clearly shows that they are of central importance in processing of proteinases in blood. However, their interaction with proteinases is so different from that of other protein proteinase inhibitors that at least to chemists their designation as inhibitors seems undesirable.

INHIBITORS WITH CLASS-SPECIFIC REACTIVE SITES

The remaining protein proteinase inhibitors differ strikingly from the macroglobulins. For an overwhelming majority of them the following three statements are justified:

1. In the enzyme inhibitor complexes all enzymatic activities toward all substrates are totally abolished.
2. The inhibition is strictly competitive.
3. A particular inhibitory reactive site can inhibit only proteinases belonging to a single one of the four mechanistic classes proposed by Hartley (50).

All of these statements are frequently contradicted in the literature, mainly because of experimental errors, in our opinion (see 2). We are not aware of any well-characterized system that contradicts any of these three statements. If point 3 is granted then it is most convenient to divide the inhibitors into four major classes.

Inhibitors of Carboxyl, Metallo and Sulphydryl Proteinases

Many inhibitors of these proteinases have been reported, some of which are listed in Table 1. However, they remain an understudied group. Complete amino acid sequences are known for only two such inhibitors; three-dimensional structures for none. In spite of a few elegant studies there is no definitive proposal for the mechanism of action of any of the inhibitors listed in Table 1.

Table 1 Some inhibitors of thiol, carboxyl, and metallo-proteinases

Enzyme inhibited	Source of inhibitor	Molecular weight	Ref
Thiol proteinase			
Ficin, papain	Avian egg white	12,700	53, 54
Cathepsin B1, C	Avian egg white		55, 56
Bromelain	Pineapple stem	5,600	57
Papain, ficin, bromelain	Rat skin	74,000 13,400	58
Papain	Rabbit skin	12,500	59, 60
Papain, ficin, bromelain	<i>Bauhinia</i> seeds	24,000	61
Papain, cathepsin B1	Leucocyte cell, spleen cell	15,000	62
Papain, cathepsin B1	Bovine nasal cartilage	13,000	63
Ficin, papain	α_2 Serum protein (human)	90,000	64, 65
Human liver cathepsin B1	Immunoglobulin G		66
Carboxyl proteinase			
Pepsin	<i>Ascaris lumbricoides</i>	15,500–17,500	67–69
Cathepsin D, E	<i>Ascaris lumbricoides</i>		70
Cathepsin D	Potato	27,000	71, 72
Pepsin	<i>Bauhinia</i> seeds	24,000	61
Pepsin	<i>Scopolia japonica</i> cultured cell	4,000–6,000	73
Metallo proteinase			
Collagenase (human)	Cartilage, aorta (bovine)	11,000	74, 75
	Bovine nasal cartilage	22,000	63
	Rabbit tissues in culture	27,500–29,500	76
	Porcine aorta smooth muscle cell in culture		77
Collagenase (human gastric mucosal)	β_1 serum protein	40,000	78
Ca ²⁺ -dependent neutral proteinase	Rat liver	30,000	79
Neutral metallo proteinase	Rabbit bone		80
Carboxypeptidase A, B	Potato	4,300	81–85
Carboxypeptidase A	<i>Ascaris lumbricoides</i>	7,530	67, 86
Aminopeptidase	<i>N. crassa</i>	10,000 5,000	87, 88

Inhibitors of Serine Proteinases

The number of well-characterized inhibitors of serine proteinases far exceeds the number of described inhibitors of the three other mechanistic classes of proteinases. It is not clear whether this relative abundance is a true reflection of distribution of inhibitors in nature or only of the preferences and convenience of the biochemists who isolate them.

In contrast to other inhibitors, the mechanism of interaction of protein inhibitors of serine proteinases with their cognate enzymes is known in considerable detail. Sequencing and X-ray crystallographic studies have shown that these inhibitors are not all homologous; instead they consist of several (probably at least 10) homologous families. Surprisingly, however, most inhibitors of serine proteinases (α_2 macroglobulin excluded) interact according to the "standard mechanism" described below. This situation is analogous to the existence of at least two nonhomologous families of serine proteinases, the subtilisin and the chymotrypsin families, which do not share a common three-dimensional structure, but which hydrolyze their substrates and are inhibited by inhibitors according to the same mechanism. It is not yet possible to decide whether several heavily studied inhibitors—the rather small stable thrombin inhibitor from leeches, hirudin (89), and the class of large, relatively unstable inhibitors from mammalian plasma—also act by the standard mechanism. These are examined briefly at the end of this review.

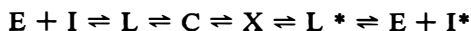
The Standard Mechanism

Inhibitors obeying this mechanism are highly specific, limited proteolysis substrates for their target enzymes. On the surface of each inhibitor molecule lies at least one (more in multiheaded inhibitors) peptide bond called the reactive site (90), which specifically interacts with the active site of the cognate enzyme. The value of k_{cat}/K_m for the hydrolysis of this peptide bond by the cognate enzyme at neutral pH is very high, 10^4 – 10^6 $\text{M}^{-1} \text{s}^{-1}$ (91, 92), compared to a typical value for normal substrates of about 10^3 $\text{M}^{-1} \text{s}^{-1}$. However, for inhibitors, the values of k_{cat} and K_m are both many orders of magnitude lower than the values for normal substrates. At typically used concentrations and neutral pH, therefore, their hydrolysis is extremely slow, and the system behaves as if it were a simple equilibrium between the enzyme and free inhibitor on the one hand and the complex on the other. The equilibrium constant for the association is extremely high (in the range of 10^7 – 10^{13} M^{-1}) (91, 2). An additional property of the inhibitory reactive sites is that their hydrolysis does not proceed to virtual completion. Instead, at neutral pH, the equilibrium constant between modified inhibitor (reactive site peptide bond hydrolyzed) and virgin inhibitor (reactive site peptide

bond intact) is near unity (92, 93–98). Since the same stable complex is formed between the enzyme and either modified or virgin inhibitor, both are thermodynamically equally strong inhibitors of the cognate enzyme (2). However, in cases so far examined the rate of complex formation from modified inhibitor and the enzyme is much lower than from virgin inhibitor and the enzyme. In a few cases, this difference is so great that it led to the assumption that the modified inhibitor was inactive, which longer incubation showed to be an error.

That the specifically hydrolyzed peptide bond in the inhibitor is the reactive site is also shown by the following experiments: (a) While in every case tried the modified inhibitor is active, specific removal of its newly formed carboxyl terminal residue (P_1) always renders it inactive (99). (b) Specific acylation of the newly formed amino terminal residue (P_1') prevents return to the virgin inhibitor and thus renders the inhibitor inactive (100). (c) Kinetically controlled dissociation of the enzyme-inhibitor complex prepared with modified inhibitor leads to the enzyme and a mixture of virgin (predominant) and modified (minor) inhibitor (101–103).

The overall mechanism of the enzyme-inhibitor interaction, including only those intermediates whose existence was definitely shown, can be written as (91, 99, 101, 103, 110, 111):



where E is the enzyme, I and I^* virgin and modified inhibitors, respectively, L and L^* are loose, noncovalent (rapidly dissociable) complexes (110) of E with I and I^* , respectively, X is the relatively long-lived intermediate in the $E + I^*$, reaction (111), and C is the stable enzyme-inhibitor complex. It is clear that more intermediates will be found as finer and finer techniques are applied to the analysis of this mechanism.

Of greatest interest to most investigators is the detailed nature of the stable complex, C. While a number of inferences can be made from thermodynamic, kinetic, and chemical modification data, the most penetrating analyses come from X-ray crystallographic examination of complexes. Three enzyme inhibitor systems have now been analyzed by X-ray crystallography: soybean trypsin inhibitor (Kunitz) with porcine trypsin (112); *Streptomyces* subtilisin inhibitor (113), S-SI, with subtilisin (114); and pancreatic trypsin inhibitor (Kunitz) (115, 116) with bovine trypsin (117, 118), bovine anhydrotypsin (119), and bovine trypsinogen in the presence and absence of isoleucylvaline (120). All of these studies show that the reactive site of the inhibitor (as previously identified by chemical studies) reacts with the active site of the enzyme in a substrate-like manner. In each case contact occurs over a small portion of the enzyme and the inhibitor, but over this

portion the fit is excellent and numerous van der Waals interactions, hydrogen bonds, and salt bridges are formed. In each case complex formation occurs with relatively small conformational change; it is predominantly a lock and key interaction, although a small conformational adaptation of the inhibitor (presumably in the $L \rightarrow C$ step) does take place.

The detailed molecular events at the active site of the enzyme and at the reactive site of the inhibitor are not as clear as some facile statements imply. Here, only the highest resolution work of the group led by Huber, summarized in several reviews (121–123), is considered. The scissile peptide bond (P_1 - P_1') of the inhibitor is intact in the complex. However, the carbonyl carbon of the reactive site peptide bond (P_1 residue) is not fully trigonal but is distorted about half way toward tetrahedral. (The older view that it was fully tetrahedral disappeared with refinement.) Since the carbon is 2.6 Å (much longer than the 1.4 Å covalent C–O bond, but shorter than a van der Waals contact) from the O^γ oxygen in the catalytic Ser¹⁹⁵ of the enzyme, it was believed that it was this oxygen that caused the tetrahedral distortion of the carbonyl carbon. This view was abandoned as it was realized that anhydrotrypsin (124) (where the catalytic Ser¹⁹⁵ residue is dehydrated to dehydroalanyl¹⁹⁵) forms stable inhibitor complexes of comparable strength (124), with essentially the same geometry (119), and with tetrahedral distortion precisely the same as in a complex with trypsin (118). Therefore the partial tetrahedral distortion is currently believed to result from the attraction between the carbonyl oxygen of the inhibitor and the oxyanion hole (125) of the enzyme, the NH's of Gly¹⁹³ and Ser¹⁹⁵. The carbonyl carbon is now poised for nucleophilic attack by oxygen O^γ of Ser¹⁹⁵, but in the stable complex the attack has not yet taken place.

In recent studies ¹³C was introduced into the carbonyl carbon of the reactive site (1-carbon of the P_1 residue) of soybean trypsin inhibitor (Kunitz) (126, 127) and of pancreatic trypsin inhibitor (Kunitz) (R. Richarz, H. Tschesche, and K. Wuthrich, personal communication). If the carbonyl carbon changed from trigonal in the free inhibitor to the completely covalent tetrahedral intermediate in the enzyme inhibitor complex (the oldest of the three models) complex formation should be accompanied by a huge (~50 ppm) upfield chemical shift. In fact, a very small (~1 ppm) downfield chemical shift is observed by all three groups that did the experiment. The complete tetrahedral intermediate is thus excluded. It is not yet clear whether these results are compatible with the X-ray results as currently interpreted.

In our opinion the X-ray crystallographic studies do not explain as yet why inhibitors are inhibitors, not just excellent substrates. This is a complex problem, since strong inhibitors of bovine trypsin are simply very good substrates for a homologous trypsin 1 from the starfish *Dermasterias im-*

bricata (92, 135). Human cationic trypsin is intermediate in behavior between these two enzymes (129). Thus, the high k_{cat}/K_m value of the reactive site of an inhibitor applies to many homologous enzymes. Some of these partition this value into very low k_{cat} and K_m , and thus are strongly inhibited; others show more conventional k_{cat} and K_m values and the inhibitor serves them as a substrate.

THE NATURE OF THE REACTIVE SITE The X-ray crystallographic studies of pancreatic trypsin inhibitor (Kunitz) (116), soybean trypsin inhibitor (Kunitz) (112), and subtilisin inhibitor, S-SI (113), permit comparison of the three-dimensional structure of these inhibitors. The overall folding of all three is dramatically different (114); they thus belong to three separate families, strongly supporting the notion that the inhibitor families arose by convergent evolution. In contrast, as seen in Table 2, the geometry of the region surrounding the reactive site (114) is virtually identical for the three inhibitors even though the sequences are quite different, and is also very similar to the "hypothetical subtilisin substrate" (128). As the experimental errors in some of these figures are large, the similarities rather than differences deserve stress at this time. The results support the notions that (a) inhibitors are predominantly very good substrates and (b) the mechanism of interaction is closely similar for enzymes belonging to the subtilisin and chymotrypsin families, and for inhibitors belonging to the various inhibitor families. The amino acid sequences surrounding the reactive sites of the various families are summarized in Table 3.

In all inhibitors that clearly obey the standard mechanism, the reactive site peptide bond is encompassed in at least one disulfide loop (90, 2), which ensures that during conversion of virgin to modified inhibitor the two peptide chains cannot dissociate. However, this requirement is not absolute since soybean trypsin inhibitor (Kunitz) with its Met⁸⁴-Leu peptide bond hydrolyzed by subtilisin is still fully inhibitory, and can be converted to

Table 2 Reactive site residues and main-chain conformations of BPTI, STI, and S-SI (114)

	P ₃		P ₂		P ₁		P ₁ '		P ₂ '	
	φ	ψ	φ	ψ	φ	ψ	φ	ψ	φ	ψ
BPTI	-86	-28	-64	152	-116	87	-135	166	-116	87
STI	-46	-21	-77	136	-89	85	(-135)	(-115)	-175	-179
S-SI	-123	141	-52	134	-92	89	-117	169	-120	78
Hypothetical substrate for subtilisin	(-157)	(172)	(-84)	(162)	(-116)	(26)	(-62)	(136)	(-112)	(-70)

Table 3 Alternative amino acid residues in the sequences surrounding the reactive site (arrow) of inhibitors

	P ₄	P ₃	P ₂	P ₁	P' ₁	P' ₂	P' ₃	P' ₄	P' ₅	P' ₆	P' ₇	P' ₈
					↓							
Pancreatic Trypsin Inhibitor (Kunitz) Family^a	...G	P	C	K	A	R	I	I	R	Y	F	Y...
		L		R	G	A	L	P	Q	F	Y	F
		R		Y	R	S	F	L	S	W	A	
		N		M	Q	Y	T	R	A	I	H	
				L		K	V	Q	L			
				-		F		S				
						H						
						M						
						P						
Pancreatic Secretory Trypsin Inhibitor (Kazal) Family^b	...G	C	P	R	I	Y	N	P	V	C...		
	V		N	K	D	L	R	L	I			
	A		T	E	E	P	H	F	L			
	L		A	D	A	F	Q	R	H			
	D		M	A	L	H	S	E				
	M		L	L	Q	D	M					
	F		S	S	N	Q	K					
				M		M	D					
				V		I						
				Q		E						
				Y								
Streptomyces Subtilisin Inhibitor Family^c	...M	C	P	M	V	Y	D	P	V	L...		
	A		T	K	Q	F				V		
Bowman-Birk Inhibitor Family^d	...A	C	T	K	S	N	P	P	Q	C...		
	M		A	R		M		G	K			
	V		A		I		A	T				
	L		L		Q							
	I		F		Y							
	S		Y									
Soybean Trypsin Inhibitor (Kunitz) Family^e	...P	S	Y	R	I	R	F	I	A	E...		

Table 3 (Continued)

	P ₄	P ₃	P ₂	P ₁	P ₁ '	P ₂ '	P ₃ '	P ₄ '	P ₅ '	P ₆ '	P ₇ '	P ₈ '
Potato Inhibitor I Family ^f	...	P	V	T	L	D	Y	R	C	N	R	...
				M		F						
						L						
Potato Inhibitor II Family ^f	...	A	S	Y	K	S	V	C	E	G	E	...
			-				-					
Alpha-1-Proteinase ^g Inhibitor Family	...	A	I	P	M	T	I	P	P	E	V	...
						S						

^aInhibitors of Kunitz family were obtained from: bovine pancreas (148); red sea turtle egg white, chelonianin (149); bovine colostrum (150); *Helix pomatia* K (151); venom of *Vipera russelli* (152); venom of *Hemachatus haemachatus* (153); venom of *Naja nivea* (153); various toxins from the venom of *Dendroaspis polylepis polylepis* toxin E (154); toxin K (155), and toxin I (156); sea anemone, *Anemonia sulcata* (157); β chain of β_1 -bungarotoxin (146); and bovine and human inter- α -trypsin inhibitor (158, 159).

^bInhibitors of Kazal family were obtained from: bovine pancreas (138); ovine pancreas (160); porcine pancreas (161); canine pancreas (K. Hochstrasser, unpublished); human pancreas (162); turkey pancreas (163); guinea pig seminal plasma (166); leech (*Hirudo medicinalis*) (141); canine submandibular glands (164, 165); ovomucoids (avian egg white) (44); Japanese quail and chicken ovomucoid (egg white) and chicken α_2 -proteinase inhibitor (serum) (39); and porcine seminal plasma (166).

^cInhibitors of *Streptomyces subtilisin* inhibitor family were obtained from: *Streptomyces albobrisesolus* (137) and *Streptomyces antifibrinolyticus* (136).

^dInhibitors of Bowman-Birk inhibitor family were obtained from: lima bean IV, IV' (168, 169); soybean Bowman-Birk inhibitor (170); soybean D-II, E-I (171); soybean C-II (172); garden bean II and II' (173); chick pea (174); adzuki bean I (175); and *Vicia angustifolia* (176).

^eInhibitor of soybean trypsin inhibitor (Kunitz) family was obtained from soybeans (167).

^fInhibitors of potato inhibitors I and II families were obtained from: var. Russet Burbank (177, 178); var. Dan Shaku Imo (179-181).

^gInhibitor of α_1 -proteinase inhibitor family was obtained from human plasma (182, 183).

modified inhibitor (Arg⁶³-Ile reactive site hydrolyzed) without loss of the 64-84 fragment (129). Clearly, strong noncovalent bonding can substitute for a disulfide bridge. Thus, eglin (130) and α_1 proteinase inhibitor (131), which are devoid of intramolecular disulfide bridges, could still obey the standard mechanism. Alternatively, the standard mechanism may have to be modified to allow for release of an inhibitor fragment from complexes formed with these inhibitors.

The reactive site residue, P₁, generally corresponds to the specificity of the cognate enzyme. Thus inhibitors with P₁ Lys and Arg tend to inhibit trypsin and trypsin-like enzymes (2), those with P₁ Tyr, Phe, Trp (artificial only), Leu and Met inhibit chymotrypsin and chymotrypsin-like enzymes, and those with P₁ Ala and Ser inhibit elastase-like enzymes. Those that inhibit chymotrypsin often, but not always, inhibit subtilisin. These, however, are

very rough rules. In the Kazal secretory inhibitor family, inhibitors with P_1 Leu and Met are strong inhibitors of elastase (44); in the Bowman-Birk family elastase is inhibited with P_1 Ala, but not with P_1 Leu (37, 132). In many instances strong inhibitors of trypsin with P_1 Arg and Lys inhibit chymotrypsin on the same reactive site (37, 133, 134). In other cases related inhibitors do not inhibit chymotrypsin, and failure to inhibit is not yet fully predictable. The most striking feature of P_1 specificity is that exchange of Lys for Arg at this position, either by actual mutation or by semisynthetic replacement (104, 107), leaves the inhibitor specificity and strength approximately the same. The exchange of Lys or Arg for a chymotrypsin specific residue generally changes the inhibitor from a good trypsin inhibitor to a good chymotrypsin inhibitor (105, 107). An odd exception is the rather strange behavior of Phe⁶³ soybean trypsin inhibitor (Kunitz) (105, 106).

The ability to tolerate a synthetic or mutational replacement of the P_1 residue and still retain inhibitory activity, either with retention of original specificity (e.g. after Arg $\rightarrow$ Lys or Leu $\rightarrow$ Met exchanges) or with dramatic change in specificity (e.g. tryptic to chymotryptic after Arg $\rightleftharpoons$ Trp exchanges), is peculiar to proteinase inhibitors. In most biologically active proteins the replacement of active site residues, even by closely related ones, leads to a complete loss or dramatic decrease in activity. We believe that largely because of this chemical difference the evolutionary behavior of most proteins and inhibitors differ. In most proteins, active site residues are stubbornly conserved, whereas in every inhibitor family when enough examples are known, there is appreciable variation in P_1 residues. This occurs often even between orthologous proteins from closely related species [e.g. plasminostreptin (Lys) (136) and S-SI (Met) (137); bovine (Arg) (138) and porcine (139) (Lys) pancreatic secretory inhibitors (Kazal), etc]. The phenomenon takes its extreme form in third domains of avian ovomucoids (44), where the P_1 residue is the most subject to evolutionary change of all 56 residues in the domain—a phenomenon referred to as hypervariability of the reactive site. This case is only an extreme example of behavior observed in the evolution of all inhibitors.

Although contrary to our earlier opinion, the P_1' residue in most inhibitor families can tolerate a broad range of residues, there are two exceptions: (a) It is generally agreed that serine proteinases seldom hydrolyze bonds with P_1' Pro. Apparently, inhibitors with P_1' Pro are not active. Silver pheasant ovomucoid with . . . Cys Asn Lys $\downarrow$ Ala . . . reactive site sequence inhibits trypsin; highly homologous golden pheasant ovomucoid with . . . Cys Asn Lys $\downarrow$ Pro . . . does not (W. J. Kohr and M. Laskowski, Jr., unpublished). Except for this case P_1' Pro is not observed. (b) There is stubborn conservation of P_1' Ser in all known Bowman-Birk inhibitors. Semisynthetic substitution study (109) showed that Ser in this position was

better than any other amino acid residue whose substitution was attempted. Rather surprisingly a synthetic variant of pancreatic trypsin inhibitor (Kunitz) with ... Cys Lys ↓ Phe ... reactive site (140) is a much weaker inhibitor than the natural ... Cys Lys ↓ Ala ... inhibitor.

Generalizations about other positions near the reactive sites are harder to draw. In all active inhibitors (note that the tables include reactive site sequences of some inactive molecules clearly homologous to active inhibitors) the residues in positions P_2 are neutral, consistent with their burial in the enzyme-inhibitor interface. Where P_2 is not required to be Cys, it is quite often Thr or a single base mutant of this amino acid. Contrary to our earlier claim, Pro is not required in the neighborhood of the reactive site as bdellin (141), a trypsin inhibitor from leeches, has no Pro residues. However, apparently at several positions (P_3 , P_2 , and probably P_3' and P_4') Pro helps to ensure the proper reactive site geometry, even though this can be obtained without Pro (see Tables 2 and 3).

Semisynthetic modifications also led us to the conclusion (108, 142) that in soybean trypsin inhibitor (Kunitz) removal of the amino acid residue P_1' leads to inactivation of the inhibitor, presumably because reformation of the P_1 - P_2' peptide bond is not possible. Similarly, insertion of an additional residue between P_1 and P_1' also led to inactivation even though all the peptide bonds were reformed, presumably because the required reactive site geometry (Table 2) was destroyed.

MULTIPLE REACTIVE SITES ON A SINGLE POLYPEPTIDE CHAIN

Where two (or more) enzymes can be inhibited by the same inhibitor molecule, two operationally distinguishable situations can arise. The enzymes can compete for the inhibitor (such inhibitors are unfortunately referred to as polyvalent) or they can be inhibited simultaneously and independently. Both situations are quite common. In the competitive case we can envisage either inhibition on the same reactive site or inhibition on two distinct, overlapping reactive sites where steric hindrance prevents association with both enzymes. Overlapping reactive sites seem quite plausible, but years of searching have yielded no convincing examples of such inhibitors of known amino acid sequence. However, such an explanation continues to be attractive for many less well-known inhibitors.

Numerous cases of independent inhibition of several enzymes are known. Functionally, inhibitors are classified as single-headed if they have only one active reactive site, double-headed if they have two and so on (143). Multi-headedness results from the presence of several active sites on the same molecule, and is most commonly achieved in protein chemistry by noncovalent association of several peptide chains each with one active site. Multi-headedness is achieved this way for at least two inhibitors. The subtilisin

inhibitor, S-SI, is a strong noncovalent dimer (144) that inhibits two molecules of subtilisin without dissociation. Similarly, potato inhibitor I is a tetramer which inhibits four molecules of chymotrypsin (145). While several other inhibitors associate (most notably Bowman-Birk inhibitors), they apparently dissociate during complex formation. Multiheadedness also results from the presence of two separate chains linked by one or more disulfide bridges. Turtle egg white inhibitor testudin [a Kazal inhibitor (I. Kato and M. Laskowski, Jr., unpublished results)] and β_1 bungarotoxin (146) (a Kunitz PTI type inactive inhibitor) are partial examples of such inhibitors.

The most common method of achieving multiheadedness is by gene duplication. The first such case discovered was lima bean inhibitor (147), a member of the Bowman-Birk inhibitor family. Inhibitors of this family consist of two homology regions, each with one reactive site in a homologous position. Even more striking examples are found in the Kunitz and Kazal families of inhibitors. In the Kunitz family, the COOH terminus of the inter- α -trypsin inhibitor consists of two tandem domains connected only by a short connecting peptide. Dog submandibular inhibitor consists of two, all avian ovomucoids of three, and avian ovinhibitors (also inhibitors from avian serum) of six tandem Kazal type domains. There is weaker evidence for several other inhibitors with multiple tandem homologous domains. In most cases, domains can be separated by cleavage of a single peptide bond in the connecting peptide and are independently active.

The tendency to form multiheaded inhibitors by gene elongation through partial or complete duplication (often repeated several times) is a general tendency of protein proteinase inhibitors and stands in reasonably sharp contrast with most biologically active proteins where a single active site in a single polypeptide chain is most common. The minimal chemical requirement for such behavior is the dispensability of the NH_3^+ terminal group and of the COO^- terminus for activity. However, the biological driving force is, in many cases, obscure. It is clear for the plasma inhibitors—the inter- α -trypsin inhibitor (158, 159) in mammalian plasma and the six Kazal domain inhibitor (39) in chicken plasma—that formation of multiple domains raises the molecular weight and prevents excretion and yet does not require the use of a large number of amino acids per reactive site. In fact, it is surprising that several mammalian plasma inhibitors (e.g. α_1 proteinase inhibitor, α_1 antichymotrypsin) do not have multiple reactive sites. No similar rationalizations are available for the Bowman-Birk inhibitors, for ovomucoids, and for dog submandibular inhibitor where the physiological functions are unknown.

Gene elongation by duplication is not the only way in which inhibitors become multiheaded. Red sea turtle inhibitor, chelonianin (Kunitz type),

(149) consists of two domains belonging to two different inhibitor families. Thus, elongation by gene fusion is another possible mechanism. The extent to which this mechanism is used is not yet known.

The Inhibitor Families

The number of known and partially characterized inhibitors of serine proteinases is enormous. They can be classified into at least 10 and probably more families. To conserve space we adopt the conservative view and describe only families for which at least one sequence has been unambiguously determined, where the reactive site has been cleanly assigned, and where there is clear evidence that the standard mechanism is obeyed. The tentative list of such families is presented in Table 4.

While the major criterion for establishing a family is extensive homology among its members, the easiest family characteristics to appreciate at a glance are the topological relationships between the disulfide bridges and the location of the reactive site. These are presented in Figure 1. In each family (except possibly potato II and *Ascaris* trypsin inhibitor) the positions of all intrachain disulfide bridges are completely conserved. A few inhibitors (or inhibitor homologues) have additional interchain disulfide bridges (e.g. testudin and β_1 bungarotoxin).

THE PANCREATIC TRYPSIN INHIBITOR (KUNITZ) FAMILY This family is named after the first inhibitor to be isolated in crystalline form (184), the first for which the typical 1 : 1 enzyme-inhibitor stoichiometry was determined (184, 185), the first for which reversibility was demonstrated (184), the first to be sequenced (148), and the first whose three-dimensional structure was determined crystallographically both for the free inhibitor and for the enzyme-inhibitor complex (115–119). This was the first, and to date the only inhibitor to be synthesized in the laboratory (140, 186, 187). Since it is small (58 amino acid residues), soluble, stable, and its sequence

Table 4 Families of protein inhibitors that inhibit serine proteinases and obey the standard mechanism

- | | |
|-------|---|
| I. | Bovine pancreatic trypsin inhibitor (Kunitz) family |
| II. | Pancreatic secretory trypsin inhibitor (Kazal) family |
| III. | <i>Streptomyces</i> subtilisin inhibitor family |
| IV. | Soybean trypsin inhibitor (Kunitz) family |
| V. | Soybean proteinase inhibitor (Bowman-Birk) family |
| VI. | Potato I inhibitor family |
| VII. | Potato II inhibitor family ^a |
| VIII. | <i>Ascaris</i> trypsin inhibitor family ^a |
| IX. | Other families |

^a Information on families VII and VIII is marginal. It may be that establishment of these families is not yet justified.

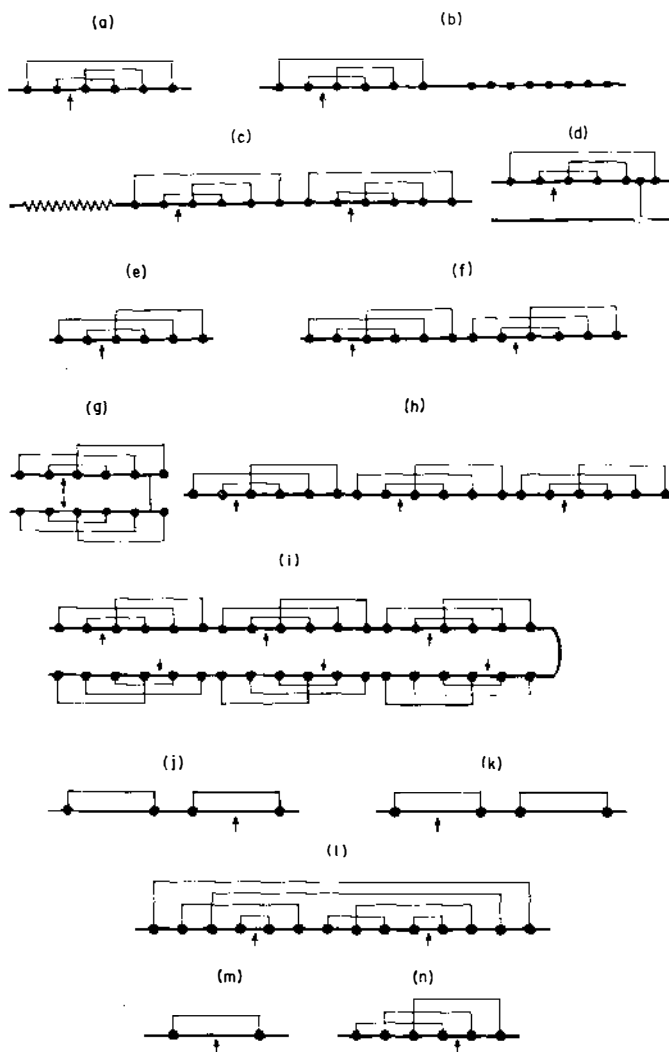


Figure 1 Topological structures of proteinase inhibitors. Solid circle and arrow indicate, respectively, a half cystine residue and a reactive site. Inhibitors of the bovine pancreatic trypsin inhibitor (Kunitz) family obtained from: *a*, bovine pancreas, Kunitz, (148); *b*, red sea turtle egg white, chelonianin (149); *c*, human inter- α -trypsin inhibitor (158, 159); *d*, B-chain of β_1 bungarotoxin (146).

Inhibitors of pancreatic secretory trypsin inhibitor (Kazal) family obtained from: *e*, bovine pancreas (138); *f*, canine submandibular glands (164, 165); *g*, red sea turtle egg white, *testudin* (Kato, I., Laskowski, M., Jr., unpublished results); *h*, avian egg white, ovomucoid (44); *i*, avian egg white ovoinhibitor (39).

Members of other inhibitor families obtained from: *j*, *Streptomyces albobrisesolus* (137); *k*, soybean, Kunitz (167); *l*, lima bean, Bowman-Birk (168, 169); *m*, potato, I (177, 178); *n*, potato II (179-181).

and three-dimensional structure are known, it became a favorite for investigations by physical protein chemists, rivalling serum albumin and ribonuclease. It has been the subject of extensive studies on protein folding (188) (time course of formation of disulfide bridges), of extensive ^1H and ^{13}C nuclear magnetic resonance studies (189–191), and it is one of the favorite proteins for testing algorithms relating sequence to three-dimensional structure (192–198). It is worth pointing out that many other protein proteinase inhibitors share all the desirable properties of pancreatic trypsin inhibitor (Kunitz). In spite of extensive studies, the physiological function of this inhibitor is not known. It is present in virtually all bovine organs (38) (liver, lungs, parotid glands, spleen), but appears to be restricted only to bovids and caprids (199).

However, both cows and pigs have a homologous inhibitor, the colostrum trypsin inhibitor (200, 201). The bovine colostrum inhibitor was sequenced (150); it is a glycoprotein with a glycosylated Asn residue (202). Its physiological function is controversial; one suggestion is that it protects colostrum antibodies against tryptic digestion in the newborn (203). Evidence both for and against this suggestion has been adduced (204–206).

The inter- α -trypsin inhibitor in mammalian serum is a huge molecule with a mol wt of 160,000. Treatment with an excess of trypsin liberates from it a COOH-terminal fragment of mol wt 14,000 (207). This fragment was sequenced from both bovine and human inter- α -trypsin inhibitor (158, 159). It consists of two tandem domains, each homologous to bovine pancreatic trypsin inhibitor (Kunitz), connected by a short connecting peptide. The first of these domains (the penultimate domain of inter- α -trypsin inhibitor) has Met (human) or Leu (bovine) as a P_1 residue. Surprisingly, it does not inhibit any of the enzymes tested against it. The second domain (the ultimate domain in inter- α -trypsin inhibitor) has Arg as P_1 in both human and bovine material. It inhibits trypsin. The sequence preceding the two COOH terminal domains of inter- α -trypsin inhibitor is not known; it may contain several more Kunitz type domains. Fragmentation of inter- α -trypsin inhibitor also occurs *in vivo*. The fragments in serum are identified as acid-stable serum proteinase inhibitors. Because of their low molecular weights they are rapidly excreted and can be isolated from urine, when they are called mingin (208). Since many proteinase inhibitors consist of several independent domains connected by connecting peptides, these findings, so far unique, suggest that many smaller inhibitors will turn out to be fragments of larger ones. The presence of bovine pancreatic trypsin inhibitor (Kunitz), bovine colostrum inhibitor, and at least two Kunitz-type domains in inter- α -trypsin inhibitor shows that the bovine genome codes for at least four Kunitz type inhibitor domains.

So far, no Kunitz type inhibitors have been reported from birds. However, snake venoms proved an unexpectedly rich source. Four different

single domain Kunitz type inhibitors (toxins I, K, B, and E) were isolated, and three of them (I, K, and E) were sequenced from the venom of the black mamba (154–156). The striking property of these is that toxin K (P_1 Lys) and toxin I (P_1 Tyr) do not inhibit trypsin and chymotrypsin, respectively, but toxin E (P_1 Lys) is a strong inhibitor of trypsin. Several Kunitz type inhibitors of trypsin were isolated and sequenced from venoms of several snakes (152, 153). β_1 Bungarotoxin (146) consists of two polypeptide chains held together by an interchain disulfide bridge. Chain A of 120 residues is homologous to phospholipase A, but chain B is homologous to Kunitz type inhibitors. However, chain B differs from the other members of the Kunitz group in two important ways. The residue corresponding to reactive site P_1 is deleted, thus suggesting that this is not an inhibitor, and there are seven (rather than the common six) half-cystine residues. The seventh residue is presumably responsible for the interchain disulfide. The β_1 bungarotoxin structure shows that Kunitz type inhibitor sequences may be covalently associated with protein sequences other than multiple Kunitz domains.

A much clearer example of this phenomenon is shown by the Kunitz type inhibitor from red sea turtle egg white, chelonianin (149). This inhibitor inhibits trypsin (presumably in its first domain where P_1 is Lys) and subtilisin (presumably on the second domain), independently. The first domain is closely homologous to Kunitz type inhibitors, while the second domain is not homologous to any of the inhibitor families described in this review and therefore presumably represents a new inhibitor family. Even though the subtilisin reactive site has not yet been assigned, it seems likely that the chelonianin is the first and, so far, the only example of a multidomain inhibitor, where the domains stem from different inhibitor families.

The Kunitz type inhibitors are not confined to vertebrates: several closely related inhibitors were isolated from the mucus of the garden snail (209, 210). Some of these have P_1 Arg, others P_1 Lys. One of them, K (P_1 Lys, 151), was sequenced. An inhibitor of the Kunitz class in sea anemone (211, 212) has also been partially sequenced (157). The gene coding for Kunitz type inhibitors is thus both very old and very widely distributed. It appears to have been present at the inception of the radiation of multicellular animals; in many animals it is present in several paralogous copies.

PANCREATIC SECRETORY TRYPSIN INHIBITOR (KAZAL) FAMILY

The family is named after bovine pancreatic secretory trypsin inhibitor (Kazal) (213) which was found in all the vertebrates examined. The bovine (138), ovine (160), porcine (161), canine (K. Hochstrasser, personal communication), human (162), and turkey (163) inhibitors were sequenced. The inhibitors are stored in zymogen granules and secreted with the zymogens in the pancreatic juice. Consistent with their function is the clear specificity for inhibition of trypsin, failure to interact with trypsinogen, failure to

inhibit enterokinase (if they did, trypsinogen could not be activated in the duodenum) and other pancreatic serine proteinases of its own species (the inhibitor would then be diverted from its essential target, trypsin). These properties contrast with those of the pancreatic trypsin inhibitor (Kunitz). A large number of low-molecular-weight inhibitors were isolated from the sperm, seminal plasma, and seminal vesicles of various mammals (32). Most of these are referred to as acrosin inhibitors, since acrosin is a trypsin-like enzyme located in sperm acrosomes (215, 216). However, the physiological function of these inhibitors is far from clear. Two such inhibitors were sequenced; one from porcine seminal plasma completely and one from guinea pig seminal vesicles partially (166). Both are single domain inhibitors that are clearly homologous, but not identical, to Kazal pancreatic secretory inhibitors. It is highly likely that several other unsequenced inhibitors of this type also belong to the Kazal family on the basis of their amino acid analysis.

A double-headed inhibitor of trypsin (on one site) and of chymotrypsin, subtilisin, and elastase (on the other site) is present in extremely large amounts in dog submandibular glands (164, 214). This inhibitor consists of two tandem Kazal type domains, connected by a short connecting peptide (165). The first trypsin-inhibiting domain has P₁ Arg, the second P₁ Met. Dog seminal plasma inhibitor probably also exists and is another member of this class.

Ovomucoids, the major glycoprotein proteinase inhibitors from avian egg whites, have long been known, and chicken ovomucoid, a single-headed inhibitor of trypsin, has been widely studied (217). However, examination of inhibitory properties of ovomucoids from various birds produced a major surprise (218, 219). Some were single-headed (e.g. chicken), some double-headed (e.g. turkey, which has an independent site for trypsin and another for chymotrypsin, subtilisin, and elastase), and yet others were triple-headed (e.g. duck, with two sites for trypsin and one for chymotrypsin, subtilisin, and elastase). Furthermore, inhibitors from closely related birds have different specificities (e.g. golden pheasant ovomucoid inhibits chymotrypsin, while chicken ovomucoid inhibits trypsin). All of these findings were easier to understand when, upon sequencing, avian ovomucoids turned out to consist of three tandem, homologous (43, 44, 220), Kazal-type domains, in which the P₁ residue of the reactive site varies widely. The connecting peptide between the second and third domains can be readily hydrolyzed, and the resultant domain III and the double domain I-II are independently active. No evidence suggests any significant noncovalent interaction between the domains. The carbohydrate in ovomucoid is attached to four (in Japanese quail and turkey) or five (in chicken) Asn residues all in Asn-X-Thr/Ser sequences. The glycosylation of the site in the third domains is partial, therefore, after enzymatic hydrolysis of the

connecting peptide, two types of third domains are isolated by gel exclusion. Both have the same amino acid sequence but one is glycosylated, and the other is not. Both have identical inhibitory properties and behave identically in the low pH transition.

Along with other proteins synthesized by the oviduct, ovomucoids became objects of considerable interest to students of gene expression, protein synthesis, and secretion (221, 222). Chicken ovomucoid has a 22 residue signal peptide, which is removed from its NH₂ terminus during synthesis. Messenger RNA for chicken ovomucoid was isolated, and cDNA copy was prepared and cloned (223, 224). The sequencing of this material as well as of the DNA gene of ovomucoid is in progress.

Aside from ovomucoid, avian egg whites contain another inhibitor of serine proteinases, ovomucoid (41). This protein is present in much smaller and more variable amounts than ovomucoid (225). It is also a glycoprotein, but with a smaller carbohydrate content than ovomucoid (226). It has been extensively studied (227–232). Sequencing (almost complete) shows that Japanese quail ovomucoid consists of six tandem, homologous, Kazal type domains (39, 45). The P₁ residues of the six domains are, in order, Arg, Arg, Arg, Tyr, Met, and Met. It is striking that these three trypsin-type reactive sites are followed by three chymotrypsin-subtilisin-elastase-type reactive sites. A proteinase inhibitor with specificity identical to that of chicken ovomucoid and with almost identical molecular properties was isolated from chicken plasma (40). It was immunologically indistinguishable from ovomucoid (40), and had the same amino acid sequence for at least the first 30 residues from its amino terminal end (45). On these bases we believe that chicken ovomucoid and chicken plasma proteinase inhibitor are identical, the slight differences arising from differences in post-translational modification. It is thus likely that the principal function of ovomucoids is to serve as the major proteinase inhibitor in avian plasma (a physiological equivalent, but not a homologue of mammalian α_1 proteinase inhibitors). On the basis of these studies birds code for at least ten paralogous Kazal type domains—six in ovomucoid, three in ovomucoid, and one in pancreatic secretory inhibitor. It is also likely that avian seminal plasma inhibitor contains a Kazal domain.

Turtle egg whites contain two proteinase inhibitors. Chelonianin, a hybrid of a Kunitz domain and another domain from a new family has already been described. The second is a two domain ovomucoid, testudin, which has two separate domains linked by one disulfide bridge.

Like Kunitz inhibitors, Kazal inhibitors are not limited to vertebrates. Crude extracts from leeches, *Hirudo medicinalis*, contain numerous trypsin inhibitors, called bdellins (233, 234). One of these, bdellin B-3, was partially sequenced. It is clearly a single-domain Kazal inhibitor (141).

STREPTOMYCES SUBTILISIN INHIBITOR FAMILY Various species of *Streptomyces* excrete into the medium protein proteinase inhibitors, with different specificities dependent upon species. One, called S-SI, from *Streptomyces albobriseolus*, inhibits subtilisin very strongly and chymotrypsin and trypsin very weakly (235, 236). It was prepared in high purity and large amount and was chosen as an object for collaborative study by several Japanese research teams. The amino acid sequence was obtained (137), the reactive site assigned, the three-dimensional structure was determined for the free inhibitor (237) and is in progress for the subtilisin-inhibitor complex (114, 238). A large number of physicochemical studies on the free inhibitor and on complex formation were made or are in progress (239–242). *Streptomyces antiplasminolyticus* excretes a potent plasmin inhibitor plasminostreptin. This inhibitor was also extensively characterized (136, 243, 244). The two inhibitors are strongly (70%) homologous. They consist of 113 and 109 residues respectively, each with two intrachain disulfide bridges. The reactive site is located in the second of these. S-SI forms strong noncovalent dimers and dimers inhibit two enzyme molecules without dissociation.

A striking observation is the pronounced sequence similarity in the reactive site region between S-SI and plasminostreptin on one hand and the Kazal family of inhibitors. In the region where it holds this similarity is sufficient to be called a homology; however, the remainder of the molecules appear unrelated. We are unable to decide whether the S-SI family and the Kazal family are products of divergent or of convergent evolution, and therefore, whether S-SI family should be a subgroup of the Kazal family or a separate family. In either case, the strong sequence similarity should make the comparison between them particularly useful in structure-to-function studies.

More *Streptomyces* inhibitors are being isolated and studied. They may prove to be as good or better than avian ovomucoid third domains as a demonstration of hypervariability of the reactive site between closely related species.

BOWMAN-BIRK INHIBITOR FAMILY These inhibitors are readily isolated from the seeds of all leguminous plants. A large number of them were briefly described (11), and a considerable number were sequenced (168–176). All of the sequenced ones consist of two tandem homology regions on the same polypeptide chain, each with a reactive site (245) (Table 3). However, a significant difference exists between the Bowman-Birk inhibitors, where homology regions are linked by interhomology region disulfide bonds (246), and the animal inhibitors from Kunitz pancreatic trypsin-inhibitor related family and Kazal secretory inhibitor family, where there are no interhomology region [inhibitor family, where there are no interhomology region (interdomain) disulfide bonds] (43). In spite of the pres-

ence of interhomology region disulfides, Ikenaka and co-workers split the two homology regions by making two specific peptide bond cleavages (CNBr and a pepsin-catalyzed cut) (247, 248). Both homology regions are separately active, but are weaker inhibitors than the parent molecule, presumably due to partial loss of rigidity. Synthetic decapeptides corresponding to the heterodetic cyclopeptide containing the reactive site were made (249). These are even more weakly inhibitory than the separated homology regions, but, nonetheless, they appear to serve as promising models for the study of the role of individual amino acid residues in inhibition.

In most Bowman-Birk type inhibitors the P_1 residue in the first (NH_2 -terminal) homology region is Lys, and trypsin is inhibited, while the P_1 residue in the second homology region is Leu, and chymotrypsin is inhibited. However, these residues vary a good deal. Particularly interesting is the soybean inhibitor D-II, in which both P_1 residues are Arg and which is double headed for trypsin. This inhibitor can be viewed as an ancestor of other inhibitors (171). Soybean inhibitor C-II (172) and garden bean inhibitor II (173), both of which have Ala and Arg as their P_1 residues, inhibit both elastase and trypsin.

Many homologous Bowman-Birk inhibitors are isolated from each species of beans or even from a single bean (171). Some of these forms appear to differ only in the length of their NH_2 -terminal sequence, but many represent significant differences in sequence. Their tendency to form homo- or hetero- dimers and trimers complicates the estimate of the number of forms present. However, Ikenaka et al isolated and sequenced three truly different inhibitors of this type from soybeans (171, 172).

SOYBEAN TRYPSIN INHIBITOR (KUNITZ) FAMILY The first plant inhibitor to be well characterized was soybean trypsin inhibitor (Kunitz). Its isolation and crystallization and that of its complex with trypsin by M. Kunitz is one of the classic achievements of inhibitor chemistry (185, 250). The sequence (167, 251, 252) and three-dimensional structure of the porcine trypsin-inhibitor complex were determined (112). This inhibitor served as the main substrate for work (largely in this laboratory) that established the standard mechanism and permitted development of semisynthetic reactive site replacement technique (104, 108). It is, therefore, surprising that this inhibitor stands rather alone—there are no reports of homologous inhibitors from the most common leguminous plants. However, a strikingly similar inhibitor from winged bean (253) was recently described, and inhibitors from rice (254) and from barley (255, 256) are probably homologous to soybean trypsin inhibitor (Kunitz).

Soybean trypsin inhibitor (Kunitz) is often confused with Bowman-Birk inhibitor, but the two are strikingly different. The polypeptide chain in single-headed Kunitz soybean inhibitor has 181 residues and only two

disulfide bridges (see Figure 1, *k*) while Bowman-Birk is typically double headed, has 7 disulfides, and only about 70 residues (Figure 1, *l*).

OTHER INHIBITOR FAMILIES We recently proposed (45) a list of inhibitor families in which we assigned the trypsin inhibitor(s) and chymotrypsin inhibitor(s) from *Ascaris* to two new separate families. It is now likely, but not certain, that one new family will suffice. *Ascaris lumbricoides* is a large parasitic worm, one type of which infests man and another pigs. These worms contain inhibitors for all of the digestive enzymes of the host (67). Pepsin and carboxypeptidase inhibitors from this source are listed in Table 1. One or more inhibitors of trypsin and several related inhibitors of chymotrypsin have been reported. The trypsin inhibitor isolated from *Ascaris* that infest pigs is an inefficient inhibitor of human trypsin, thus raising the possibility that this property may account for the inability of this species to infest humans. The sequence of this inhibitor has long been known (257), but the position of its reactive site was only recently assigned (R. J. Peanasky, W. J. Kohr, and M. Laskowski, Jr., unpublished results); in the process, the sequence was revised.

The chymotrypsin inhibitors of *Ascaris* appear to inhibit chymotrypsin, elastase, and subtilisin at the same reactive site (67). This behavior is similar to that of Kazal-type inhibitors (e.g. turkey ovomucoid third domain), but is in contrast to that of chymotryptic sites of Bowman-Birk type inhibitors, where only chymotrypsin is inhibited. Recently, both the preliminary sequence (258) of several *Ascaris* chymotrypsin inhibitors and the location of the reactive site (259) were announced.

Another extremely rich source of proteinase inhibitors are potatoes and related plants (9, 11). Many inhibitors, including carboxypeptidase inhibitor (Table 1) and several families of serine proteinase inhibitors, are present. The best understood are the potato I inhibitors, which are noncovalent tetramers of a single-chain inhibitor with only a single intrachain disulfide bridge (260, 261). Far more complex is the potato II family, where amino acid sequences have been reported for a "low-molecular-weight chymotrypsin" inhibitor (262), for an inhibitor from eggplant (263), and for inhibitory fragments of two larger serine proteinase inhibitors, IIa (264, 265) and IIb (180, 266). When all of these are aligned, homology is apparent. However, in contrast to other inhibitors, reactive sites (267, 268) do not align and there are many postulated deletions. At the present stage, therefore, the designation of a potato II family may be premature.

At least 100 small inhibitors not reported in this review have been described for serine proteinases and in some cases rather intensively studied. It seems premature to place them into families, but it is clear that new families will have to be created for some of them. The benefits in obtaining

enough information to allow for a family assignment will be quite large, since comparison of inhibitory specificities or of various thermodynamic and kinetic parameters of enzyme-inhibitor interaction is far more valid within a family.

INHIBITORS IN MAMMALIAN BLOOD PLASMA Many proteolytic processes (e.g. blood coagulation, blood clot dissolution, formation and destruction of peptide hormones) occur in blood plasma and require exquisite control. It is not surprising, therefore, that plasma contains many kinds of proteinase inhibitors—some in very high concentration. The principal ones in human plasma are listed below.

Of the inhibitors listed in Table 5, α_2 macroglobulin is a general endopeptidase trap and its postulated mechanism of action has already been discussed. Inter- α -trypsin inhibitor chain terminates in two domains belonging to the pancreatic trypsin inhibitor (Kunitz) family, the presumed reactive site lies in the COOH terminal domain. The other inhibitors listed in Table 5 do not appear to belong to any of the families discussed thus far. Furthermore, the sequence is not completely known for any of them and their mechanism of interaction with serine proteinase—in spite of a large body of literature dealing with this subject—is either unknown or highly controversial. This comparative lack of information results from the comparatively large size (required of all plasma proteins) and high lability of these inhibitors, which complicate isolation and experiments that are readily done with small stable inhibitors. As a result we resist detailed review and offer only thumbnail sketches without thorough analysis.

By far the most studied is α_1 -proteinase inhibitor (often called α_1 anti-trypsin—a bad name) because its genetic deficiency strongly predisposes individuals who are homozygous for its Z allele (in contrast to the normal M) to emphysema (270) and to liver disease. The Z inhibitor is fully active, but it is present in serum at 20% of the normal concentration. Partial sequencing shows that Z differs from M by the substitution of Lys for Glu in one of the internal sequence positions (271–273). In contrast to small inhibitors discussed earlier, α_1 -proteinase inhibitor contains no intrachain disulfide bridges (131). α_1 -Proteinase inhibitor inhibits an exceptionally broad spectrum of serine proteinases (e.g. trypsin, chymotrypsin, and elastase). A complex with one of these enzymes does not inhibit any of the others. This observation may indicate that all of the enzymes act at the same reactive site; a number of alternative explanations are also possible. If a single reactive site is accepted then good, but not compelling, evidence (182, 274) was presented that the site is a ... Met-Ser ... bond in the NH_2 -terminal portion of the molecule (the sequence surrounding the reactive site is shown in Table 3).

Table 5 Principal proteinase inhibitors in human blood plasma (269)

Name	Concentration mg/100 ml	Mol wt	Number of polypeptide chains	Heads ^a
α_1 -proteinase inhibitor	290 $\pm$ 45	52,000	1	1
α_1 antichymotrypsin	49 $\pm$ 7	69,000	1	1
inter- α -trypsin inhibitor	50	160,000	1	2
α_2 antiplasmin	7 $\pm$ 1	70,000	1	1
antithrombin III	24 $\pm$ 2	65,000	1	1
C ₁ -inactivator	24 $\pm$ 3	70,000	1	1
α_2 -macroglobulin	260 $\pm$ 70	720,000	4	1-2 or more

^a Maximal number of enzyme molecules simultaneously inhibited by one inhibitor molecule.

The most recently discovered (275-277) of the major plasma inhibitors is α_2 antiplasmin. Antiplasmin reacts slowly with several trypsin-like enzymes but its reaction with plasmin is exceptionally fast. However, its rate of association is reduced (278) to more typical values either by enzymatic removal of that portion of the plasmin heavy (A) chain containing the strong lysine binding site or by addition of moderately low concentrations of lysine or of ϵ -aminocaproic acid. The plasmin- α_2 antiplasmin-inhibitor interaction is speeded by specific interaction between the inhibitor and the strong lysine binding site on the heavy chain of plasmin (278).

Antithrombin III is the major thrombin-directed inhibitor in plasma. The thrombin-antithrombin reaction is moderately slow in absence of added heparin; addition of heparin greatly accelerates the reaction. The current consensus appears to be that it is the antithrombin-heparin complex that reacts with thrombin very rapidly (279-281). Once the enzyme-inhibitor-heparin complex is formed, most of the heparin is released, and is therefore free to combine with additional antithrombin to further "catalyze" the enzyme-inhibitor reaction (282, 283). An essentially complete amino acid sequence of antithrombin III was recently reported (284), but the thrombin reactive site cannot be easily identified. Based on only a small portion of the α_1 proteinase inhibitor sequence, it appears that antithrombin III and α_1 proteinase inhibitor may be homologous.

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